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Section of class: Monday and Wednesday 9-10:20am
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Introduction

We used several data from Federal Reserve Economic Data, including house price index, mortgage rate, household income, personal income, and building permits. The below are the specific data we used.

1. Chicago Home Price Index. The home prices base value was 100 in January 2000.
2. 30-Year Fixed Mortgage Rate average in the U.S.
3. Median Household Income (Chicago area)
4. Real Per Capita Personal Income for Chicago-Naperville-Elgin, IL-IN-WI (MSA). New Private Housing Structures Authorized by Building Permits for Chicago-Naperville-Elgin, IL-IN-WI (MSA)

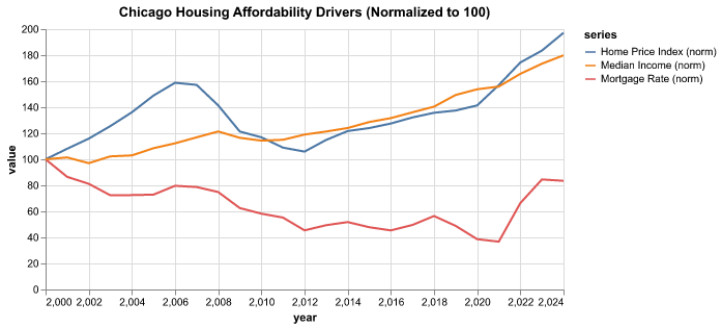
Research question, approach, and coding

This project asks how the housing price and affordability in Chicago changed, as the Fed cut or raised the rate. The approach is to find the median housing price in Chicago and the average incomeBecause median transaction prices are not available in the data so we use an anchored-price method, that is, a base home price is scaled by Chicago home price index to obtain a price proxy. We then define payment-to-income ratio (PIR): monthly mortgage payment over monthly median household income) and PTI (price-to-income ratio: anchored price over annual income) as the main affordability metrics, with API (affordability pressure index) as an auxiliary measure. Data are all pulled from FRED, annualized by resampling and merged on year from 2000. The pipeline is implemented in a ipynb notebook and we also create a Streamlit app that let users change base price, down payment and a mortgage rate shock and see how updated PIR and PTI and trend charts. All visualizations use *Altair*: normalized driver trends, time series, correlation heatmaps and a mortgage-rate vs PIR scatter plot with linear fit line.

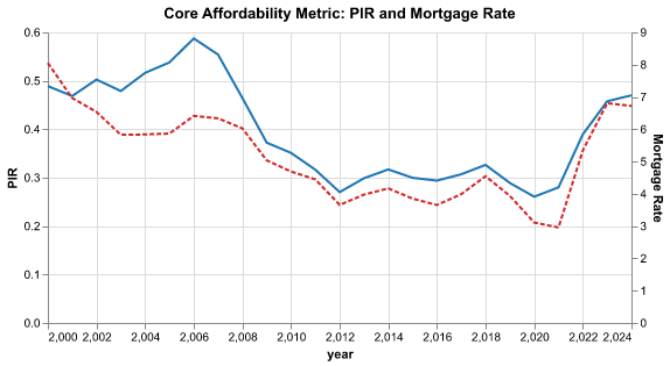
Weaknesses and difficulties

1) Trend Overview (Normalized Drivers): The “animation” is implemented with a Streamlit year slider and precomputed ±5-year windows, because Altair does not support the same kind of frame-based animation as Plotly.
2) We cannot find the data of median income of each zip code. The result can just be on the broader Chicago area

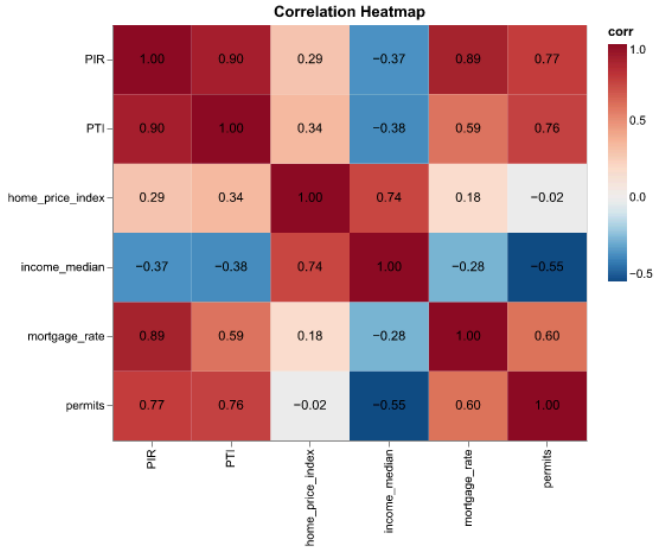
Static Plots



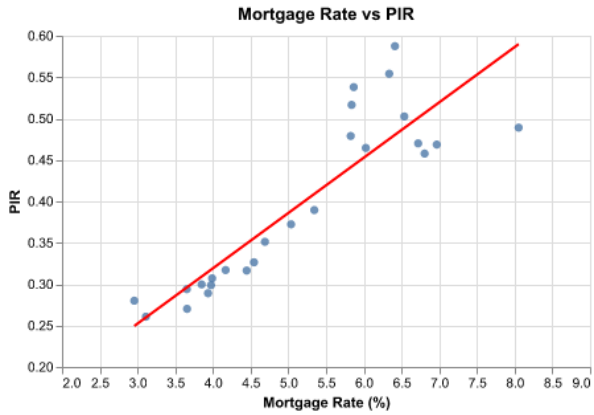
Home prices increasingly outpace income growth, and the rate rebound pushes affordability pressure upward. Home prices have risen faster than income in recent years. Mortgage rates fell first and then rebounded after 2021, creating a double pressure on financing costs.
Relate: This figure is showing that affordability pressure rises when home prices grow faster than income and mortgage rates rebound.



PIR declined during the low-rate period but rose sharply in recent years, signaling heavier monthly payment pressure. PIR has high correlation with mortgage rate
Relate: This figure directly answers our research question by tracking how household housing burden changes over time through our core affordability metric.



PIR co-moves strongly permits number and is most tightly linked to mortgage rates, while income only partially offsets pressure.
Relate: This figure relates to our research question by summarizing how affordability metrics co-move with prices, rates, income, and permits, highlighting the strongest drivers.



Mortgage rates are strongly positively associated with PIR, making financing cost a key affordability driver. Higher rates are linked to higher PIR and weaker affordability.
Relate: This figure relates to our research question by providing empirical evidence that higher mortgage rates are associated with higher PIR (worse affordability).

Streamlit

The most important plot is the dynamic plot with 3 index.

We can control the time span and midpoint of the image by adjusting the range and year, thereby observing the numerical changes over a certain period.This interactive chart lets us explore how key affordability drivers change across different time windows, helping us evaluate our research question dynamically.